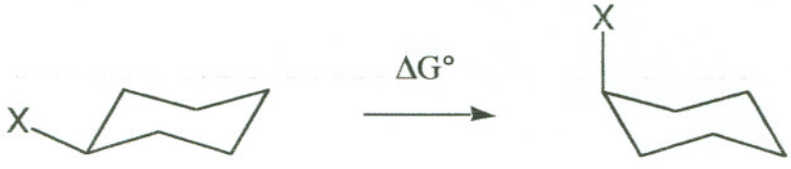
Organic Division Cumulative Exam April 1, 2005

(1) Give typical values for the bond length (to 0.1Å) and homolytic bond dissociation energies (± 5 kcal/mol) of the following bonds.

C—C	C—O	C—N
C=C	C=O	C=N
C≡C		C≡N
	C—H	
	O—H	

(2) Give the A-values (to 0.1 kcal/mol) for the following groups X.

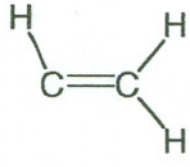


CH_3

OCH_3

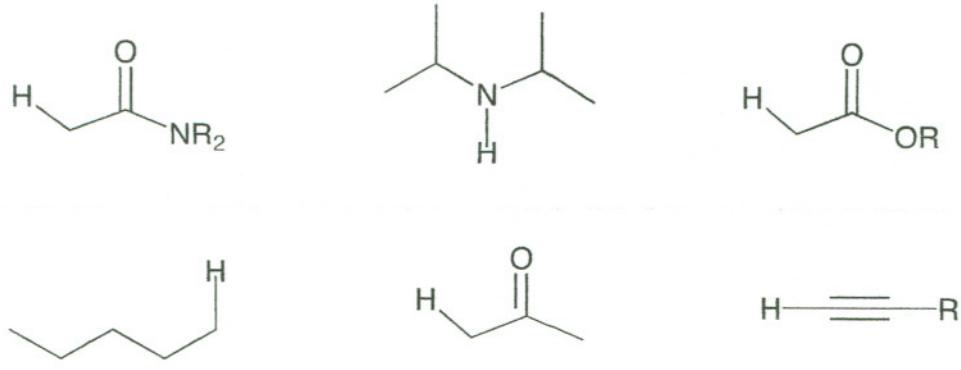
phenyl

t-butyl



C≡C—H

(3) Given two equilibrating species A and B, what energy difference in kcal/mol does there have to be between A and B to afford a 10:1 ratio of A:B at 298 K?

(4) What is the approximate vicinal coupling constant between two protons which possess an H-C-C-H dihedral angle of 180°? Of 0 °?

(5) Rank the acidity from most to least acidic and give the approximate pK_a's of the following compounds.